Hieu Nguyen

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Summary

AI engineer with 4 years of experience building production systems across retrieval, NLP, speech recognition and computer vision. Hands-on CTO work at Plenarius and Sleep Sage spans Go and Python services, mobile interfaces and tested AI workflows. Previously worked on production AI at Viettel Post and FPT Software, and researched tool-using agents at QUT. Completed a Master of Artificial Intelligence in November 2026.

Skills

- **AI/ML:** RAG, vector search and quantisation, agentic workflows, LLM orchestration, transformers, NLP, speech recognition, computer vision, anomaly detection.
- **Engineering:** Python, Go, TypeScript, PostgreSQL, pgvector, MongoDB, REST APIs, React, React Native, Expo, FastAPI, Docker, CI/CD, data pipelines.
- **Cloud & infrastructure:** Azure (Functions, Cosmos DB), AWS (EC2, S3), Hetzner, Cloudflare (WAF, R2 object storage), Caddy, systemd.

Experience

Chief Technology Officer Sep 2026 – Present
Sleep Sage

- Lead technical development of an education app that turns PDFs and text into source-grounded concepts, user-approved learning cues and private speech clips.
- Improved React Native and Expo session editing, audio playback and first-night setup, and strengthened FastAPI processing with regression tests for validation, authentication and playback.

Chief Technology Officer Apr 2026 – Present
Plenarius (formerly Calimac)

- Inherited a PHP-based platform and led its migration to Go and PostgreSQL, moving 96 server-rendered pages, 205 API paths and the recommendation system onto the new production stack with no PHP in the request path.
- Designed vector search over roughly 200,000 papers and evaluated retrieval configurations, balancing response quality, query cost and infrastructure trade-offs.
- Run the production platform on a single 8-vCPU server for roughly US$43 per month all in, tracking retrieval quality, cost per query and infrastructure trade-offs.

AI Researcher Jan 2026 – Jun 2026
Cognilaw

- Evaluated and improved a hallucination-resistant RAG pipeline for legal workflows, testing retrieval strategies, grounded generation and answer validation using retrieval and response-quality metrics.

Research Assistant Sep 2025 – Nov 2026
Queensland University of Technology

- Evaluated and engineered an agentic person re-identification system combining appearance, body composition, gait and physical attributes through confidence-weighted voting.
- Cleaned and prepared biomedical literature datasets, then fine-tuned transformer models for protein-protein interaction relation extraction.

AI Software Engineer Apr 2023 – Nov 2024
Viettel Post

- Integrated Whisper fine-tuned on Vietnamese with BERT classifiers for offensive-language detection and mood prediction, achieving 90% accuracy in production.
- Trained an Extra Trees classifier for customer churn prediction, reaching 80% accuracy.

AI Software Engineer Oct 2021 – Mar 2023
FPT Software

- Contributed to FPT's US-patented Sound AI acoustic anomaly-detection platform, deployed on a Samsung factory floor. Built its Python backend on Azure Functions, supporting model serving, CI/CD and production monitoring.

Research Intern, Computer Vision Apr 2021 – Jul 2021
Asilla

- Fine-tuned and deployed real-time human pose estimation on Jetson Nano with TensorRT, using INT8 quantisation and structured pruning to improve efficiency by 20% under edge-compute constraints.

Projects & Awards

Does Fine-Tuning Still Pay Once a Stronger Planner Starts? Preprint, Zenodo 2026
Stack: Python, AppWorld, BFCL, Agent evaluation

Co-author with Kien Nguyen Thanh at QUT of "Does Fine-Tuning Still Pay Once a Stronger Planner Starts?" (Zenodo preprint, Nov 2026). Built tool-using agent orchestration and paired replay evaluations comparing handover, advice and direct-action strategies with statistical analysis and cost-quality trade-offs.

Claude Usage Optimizer Open Source | MIT
Stack: Claude Code, multi-agent orchestration, Python

Built an efficient multi-agent orchestration kit for Claude Code: a premium model plans and reviews while cheaper worker models implement. Cut premium-model spend by up to 64% and total cost by about half versus plain Claude Code on real development tasks, at slightly faster wall-clock time. Tracked 381 real runs across 11 worker lanes, with 97% of reviewed work usable. Runs on Windows and Linux; macOS, Codex and Amp support planned.

Local AI Workflow Platform Presented | AgentCamp 2026, Brisbane Edition
Stack: Python, LangGraph, FastAPI, React Flow, Pydantic, YAML

Local-first platform for authoring, running and visualising AI workflows from declarative YAML. Each workflow is validated through a Pydantic GraphSpec contract before being compiled into runnable graphs, with approval interrupts, rollback restore and eval coverage.

Capture the Narrative 1st Place | UNSW National AI Competition
Stack: LLMs, multi-agent simulation, AWS EC2, S3

Built generative-AI agents inside a simulated election environment as an adversarial exercise in how synthetic content shifts sentiment.

SafeMind Most Viable | TANDA GenAI Hackathon
Stack: Python, Flask, OpenAI API, MongoDB, python-docx

Voice-driven workplace incident capture that pre-fills WorkCover claim forms, compressing hours of paperwork for injured workers.

Education

Master of Artificial Intelligence 2025 – 2026
Queensland University of Technology | GPA 6.538/7.0 (current)

Completed November 2026.

Bachelor of Information Technology 2018 – 2021
University of Science and Technology of Hanoi

Volunteer

Vice President Oct 2025 – Oct 2026
QUT The Emerging Coders Hub (TECH)
Ran coding and IT workshops teaching beginners to build chatbots, personal websites and AI agents, and organised career events including mock interview sessions, progressing from General Executive (Jun–Oct 2025).

Project Manager Sep 2025 – Nov 2026
QUT AI & ML Society
Led 30+ members building a predictive analytics dashboard on real insurance data for claim-delay prediction and anomaly detection, stepping up from Project Officer (Aug–Sep 2025).